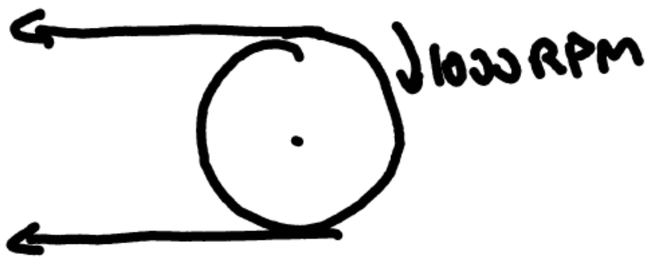
Pulley diameter: 3in r: 1.5in Center distance = 6 in

N = 1500 RPM Belt tension: 5 lbF SF = 3

1060 Hardened steel

FBD:



Unknowns: ϕ_B, F_1, F_2, μ

$T_1 = 2T - T_2$

Forces:

$T_{B1} = T_{B2} = 4$ lbF $\frac{T_1}{T_2} = e^{\mu\phi}$

$T_2 = \frac{2(4)}{e^{3\pi}+1} \Rightarrow \underline{T_2 = 2.24 \text{ lbF}}$ $T_1 = (2\times4) - 2.24$

$\Rightarrow 5.76$ lbF

Belt $\frac{S_y}{SF} = \frac{T_B}{A}$

$D = \sqrt{\frac{4 T_B SF}{T_2 S_y}}$

$D_B = \sqrt{\frac{4(4)\cdot 3}{T_2(471)}} \Rightarrow .1801$ in

$L > 3/16$ is closest

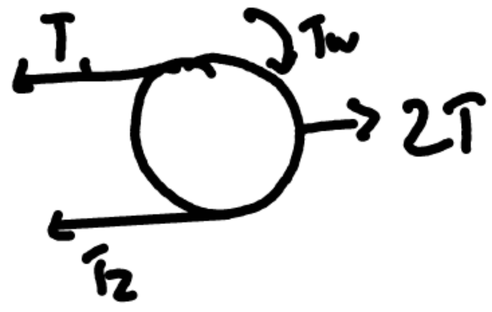
$C_B = 2L + T_2\phi_p$

McMaster belt
3044156O5

Shaft: $F_2 = 2T$ 1060 HS

RPM: 1500 rpm

FBD:



Torque $T_2\left(\frac{D_s}{2}\right)$

$$\phi_s = \sqrt[3]{\frac{16T}{\tau_{all}}}$$

$T = .577(2y)$

$T = (5.76)\left(\frac{3}{2}\right) = 8.64\ in\ lbf$

$$\phi_s = \sqrt[3]{\frac{16(8.64)}{\tau_c(56861n)}}$$

$\phi_s = .0919in$

$\approx .1 in$

Journal bearing

knowns:

$\phi_s = .1in$

Rpm = 1500

Unknowns:

P

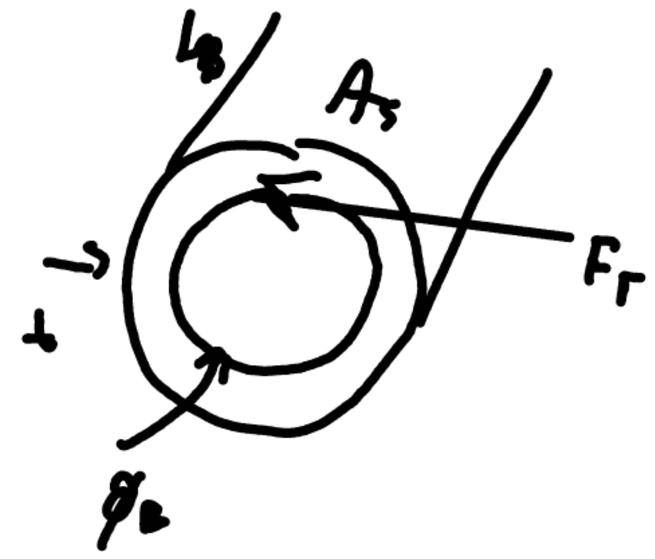
V

P_v

ϕ_B

F_r

FBD:



$$P = \frac{F_r}{A_s} = \frac{2T}{L_B \pi \phi_B'}$$

$$P = \frac{2(8.64)}{\pi(.1007)^2} \Rightarrow \boxed{543.39\ psi} \qquad .100 + 0.0007 = .1007$$

$$V = RPM\left(\frac{\pi \phi_B}{1}\right)\left(\frac{1}{12}\right) \Rightarrow 1500\left(\frac{\pi(.1007)}{1}\right)\left(\frac{1}{12}\right) \Rightarrow \boxed{39.55\ ft/min}$$

$PV = 543.34 \times 39.55 \Rightarrow \boxed{21491.07\ psi \cdot ft/mn}$

Porous Bronze

FRAME:

